

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

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Experiment 8

Effect of PCA on Regression and Classification Models

1. Aim and Objective

To study the impact of **Principal Component Analysis (PCA)** on the performance of machine learning models.

Students must:

- a) Train models in the original feature space (Without PCA)
- b) Train models in reduced feature space (With PCA)
- c) Compare performance using 5-fold cross-validation

2. Dataset Description

Dataset: **PCA_Full_Lab_Dataset_1000_Samples.csv**

Features: Academic and performance-related attributes
Regression Target: Final_Score_Regression
Classification Target: Performance_Level_Classification

3. Preprocessing Steps

1. Data Loading and Overview

The dataset was loaded into a **pandas** DataFrame. It contains information about student performance.

- Number of samples: 1000
- Number of features: 12 (after dropping target columns for X)
- Target variables:
 - **Final_Score_Regression** (for regression)
 - **Performance_Level_Classification** (for classification)

2. Target Distribution

The distribution of the `Performance_Level_Classification` target variable is as follows:

- Medium: 545
- High: 253
- Low: 202

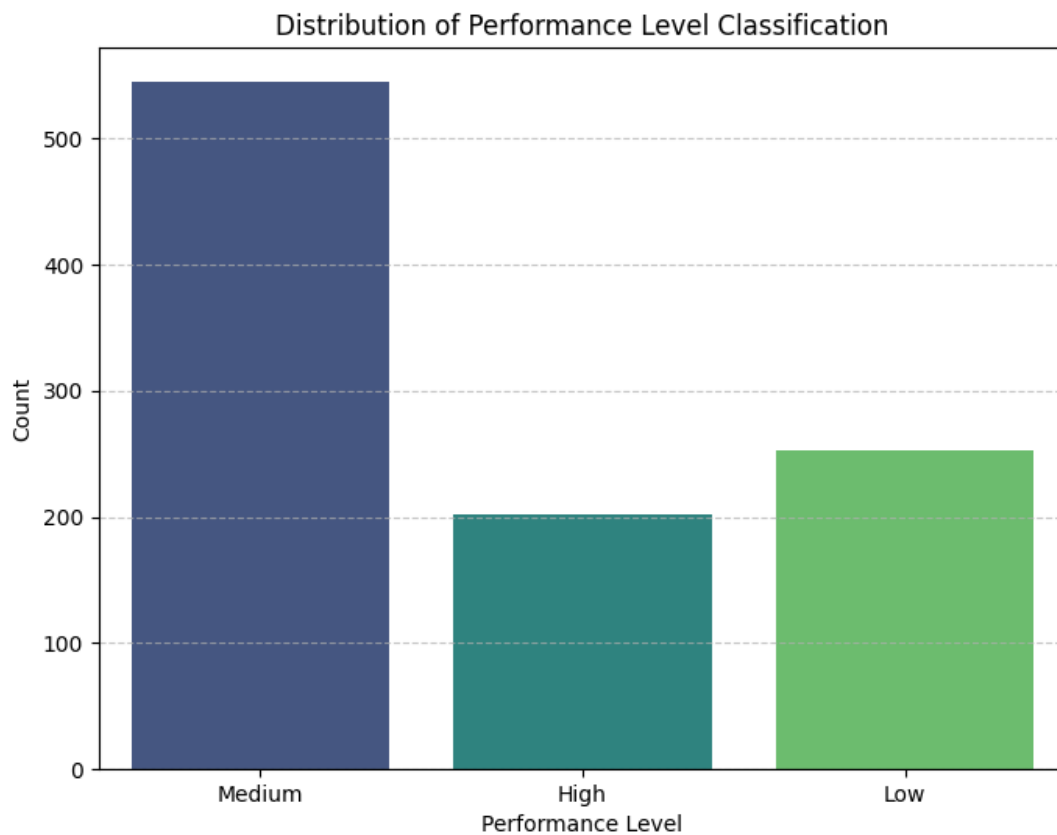


Figure 1: Target Distribution (Performance Level Classification)

This shows a slightly imbalanced distribution across the three performance levels, with 'Medium' being the most frequent class.

3. Preprocessing Steps Details

3.1 Handle Missing Values

No missing values were found in the dataset, ensuring data completeness.

3.2 Encode Categorical Target (for Classification)

The categorical target variable `Performance_Level_Classification` was encoded into numerical format using `sklearn.preprocessing.LabelEncoder`. This is necessary for machine learning algorithms that require numerical input. The mapping automatically assigns integers to the unique categories (e.g., Low, Medium, High mapped to 0, 1, 2).

3.3 Standardize Features

Features were standardized using `sklearn.preprocessing.StandardScaler`. This process scales the data such that it has a mean of 0 and a standard deviation of 1. Standardization is crucial, especially before applying PCA, as PCA is sensitive to the scale of the features.

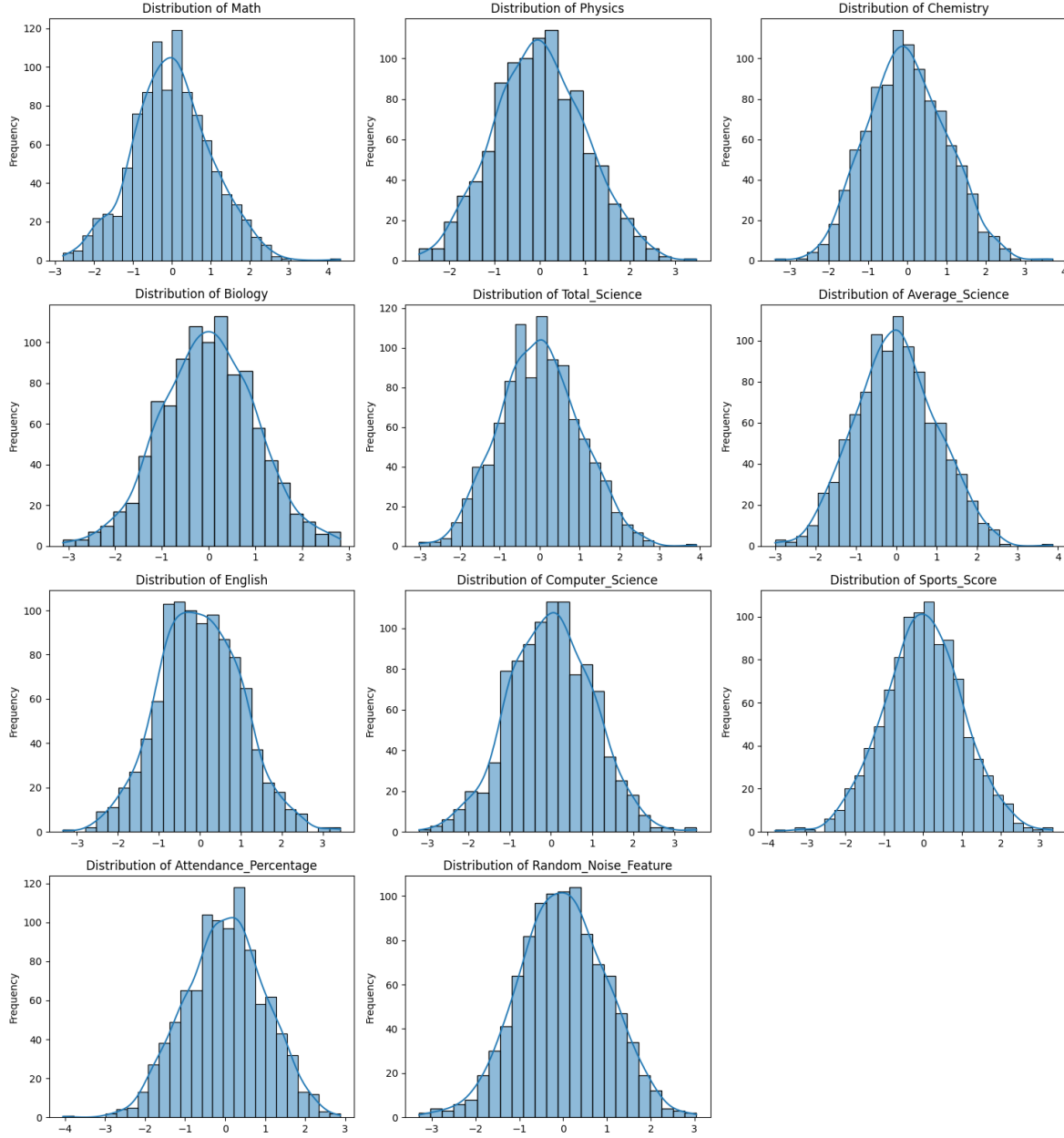


Figure 2: Feature Distribution of Academic Performance related attributes

3.4 Split Data using 5-fold Cross-Validation

A K-Fold cross-validation strategy was defined with `n_splits=5`, `shuffle=True`, and `random_state=42`. This ensures that the data is split into 5 folds, randomly shuffled, and reproducible. This approach is used for robust model evaluation and hyperparameter tuning.

4. Implementation Details

4.1 Feature Scaling

Feature scaling was performed using `StandardScaler` to normalize the data. This process ensures that features have a mean of 0 and a standard deviation of 1, which is crucial for algorithms sensitive to feature magnitudes, such as PCA and SVM.

4.2 Dimensionality Reduction with PCA

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset while retaining a significant portion of its variance. The original features were transformed into a new set of uncorrelated principal components.

- **Components Chosen:** PCA was configured to retain 95% of the variance, resulting in 7 principal components.
- **Explained Variance:** The selected components explain 95.89% of the total variance.
- **Justification:** This threshold reduces feature space while preserving most of the information. The Scree Plot (`Screeplot.png`) supports this choice, showing an elbow around the 7th component.

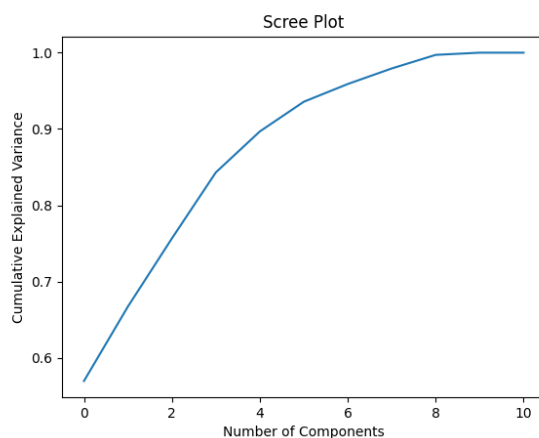


Figure 3: Scree Plot

Table 1: PCA Summary

Components Chosen	Explained Variance (%)	Justification
7	95.89	Retained components to explain 95% variance

4.3 Regression Models Implementation

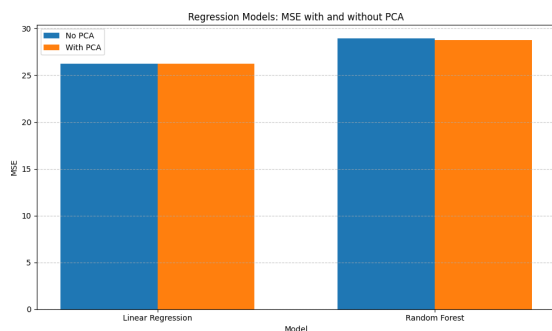
Two regression models were evaluated: Linear Regression and Random Forest Regressor. Both were trained with and without PCA to assess the impact of dimensionality reduction.

4.3.1 Linear Regression

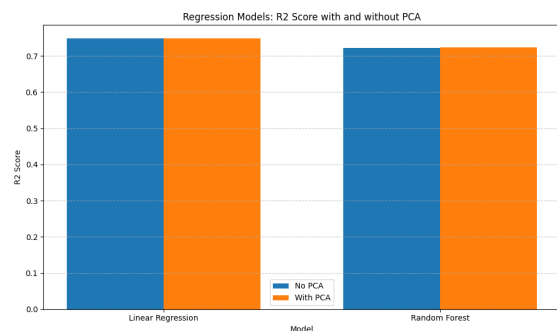
- **Model:** `LinearRegression()` from `sklearn.linear_model`
- **Hyperparameter Tuning:** Not performed (default parameters used)
- **Performance (5-Fold CV):**
 - MSE (No PCA): 26.23
 - MSE (With PCA): 26.23
 - R^2 (No PCA): 0.7479
 - R^2 (With PCA): 0.7481

4.3.2 Random Forest Regressor

- **Model:** `RandomForestRegressor()` from `sklearn.ensemble`
- **Parameters:** `n_estimators=100`, `max_depth=10`, `random_state=42`
- **Hyperparameter Tuning:** Not performed
- **Performance (5-Fold CV):**
 - MSE (No PCA): 28.92
 - MSE (With PCA): 28.76
 - R^2 (No PCA): 0.7225
 - R^2 (With PCA): 0.7241



(a) Regression MSE Comparison



(b) Regression R2 Comparison.

Figure 4: Bar Charts for Regression Evaluation Metrics

Table 2: 5-Fold CV Results – Regression

Model		Metric	Fold Avg (No-PCA)	Fold Avg (With-PCA)	Observation
Linear Regression		MSE	26.23	26.22	Very slight improvement with PCA
Linear Regression		R^2	0.7479	0.7481	Very slight improvement with PCA
Random Forest		MSE	28.92	28.76	Slight improvement with PCA
Random Forest		R^2	0.7225	0.7241	Slight improvement with PCA

Regression Results Table

4.4 Classification Models Implementation

Two classification models were evaluated: Logistic Regression and Support Vector Machine (SVM). Both were trained with and without PCA.

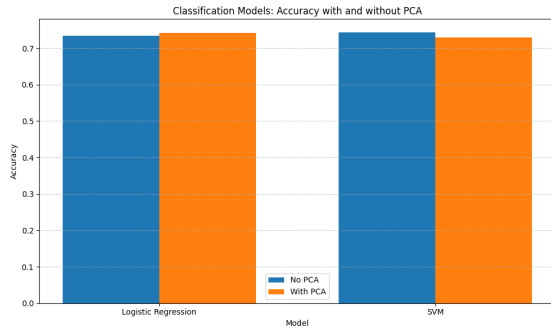
4.4.1 Logistic Regression

- **Model:** `LogisticRegression()` from `sklearn.linear_model`
- **Parameter:** `max_iter=1000`
- **Hyperparameter Tuning:** Performed using `GridSearchCV`
 - Cross-validation: 5-fold (`KFold(n_splits=5, shuffle=True, random_state=42)`)
 - Scoring: Accuracy
 - Parameter Grid: `C = [0.001, 0.01, 0.1, 1, 10, 100]`
 - Best Parameter: `C = 0.01`
 - Best CV Accuracy: 0.7480
- **Performance (5-Fold CV):**
 - Accuracy (No PCA): 0.734
 - Accuracy (With PCA): 0.742
 - F1-score (No PCA): 0.731
 - F1-score (With PCA): 0.740

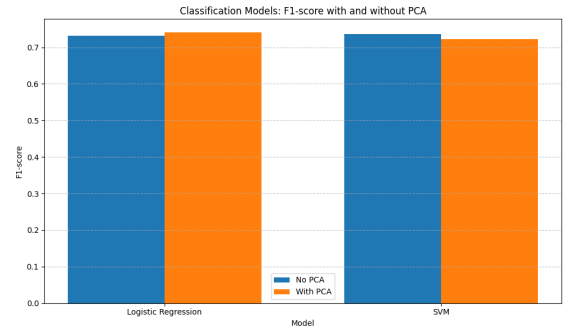
4.4.2 Support Vector Machine (SVM)

- **Model:** `SVC()` from `sklearn.svm`
- **Kernel:** `rbf`
- **Hyperparameter Tuning:** Not performed

- **Performance (5-Fold CV):**
 - Accuracy (No PCA): 0.743
 - Accuracy (With PCA): 0.730
 - F1-score (No PCA): 0.736
 - F1-score (With PCA): 0.722



(a) Classification Accuracy Comparison



(b) Classification F1 Comparison

Figure 5: Bar Charts for Classification Evaluation Metrics

Classification Results Table

Table 3: 5-Fold CV Results – Classification

Model	Metric	Fold Avg (No-PCA)	Fold Avg (With-PCA)	Observation
Logistic Regression	Accuracy	0.734	0.742	PCA slightly improved accuracy
Logistic Regression	F1-score	0.731	0.740	PCA slightly improved F1-score
SVM	Accuracy	0.743	0.730	Slight decrease with PCA
SVM	F1-score	0.736	0.722	Slight decrease with PCA

Analysis

Effect of PCA on Linear Regression

Performance: PCA resulted in a very slight improvement. The MSE remained nearly the same (26.23), while R^2 increased from 0.7479 to 0.7481.

Reason: Linear models are sensitive to multicollinearity and noise. PCA transforms correlated features into uncorrelated components, improving stability and slightly enhancing performance.

Effect of PCA on Random Forest

Performance: PCA led to a small improvement. The MSE decreased from 28.92 to 28.76, and R^2 increased from 0.7225 to 0.7241.

Impact of PCA on Classification Models

Logistic Regression: Accuracy improved from 0.734 to 0.742 and F1-score from 0.731 to 0.740.

SVM: Accuracy decreased from 0.743 to 0.730 and F1-score from 0.736 to 0.722.

Conclusion: Logistic Regression benefited more from PCA, while SVM performance slightly declined.

Variance Across Folds

Observation: Only average cross-validation scores are provided. Since variance or standard deviation is not available, it cannot be determined whether PCA reduced variance across folds.

Why Linear Models Benefit More from PCA

Linear Models: Models like Linear Regression and Logistic Regression are sensitive to feature correlation. PCA reduces dimensionality and removes multicollinearity, making the data more suitable for these models.

Ensemble Models: Models like Random Forest are less sensitive to correlated features and high dimensionality. They perform implicit feature selection and capture complex relationships. PCA may remove useful information, leading to smaller improvements or slight performance drops.

Learning Outcomes

- PCA reduces multicollinearity
- PCA helps linear models more
- Tree-based models may not benefit much
- PCA reduces noise
- Dimensionality reduction can improve generalization